# Chapter 2: The Production Possibilities Frontier & Gains from Trade

Principles of Microeconomics Curriculum

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**Abstract**

This chapter provides a rigorous, pedagogical treatment of Chapter 2: The Production Possibilities Frontier & Gains from Trade, benchmarked to standard principles curriculum.

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## 1 The Production Possibilities Frontier (PPF) Model

The PPF illustrates the boundary of maximum output combinations achievable under fixed technology and full factor utilization.

$$MRT = -\frac{\Delta Y}{\Delta X} = \left|\frac{dY}{dX}\right| = \frac{\text{Quantity of Good } Y \text{ Forgone}}{\text{Quantity of Good } X \text{ Gained}} \tag{1}$$

## 2 Law of Increasing Opportunity Cost

Concave curvature reflects imperfect factor adaptability. As resources shift toward one sector, inputs progressively less suited to that industry must be utilized, driving opportunity costs upward.

## 3 Ricardian Comparative Advantage

Comparative advantage is determined strictly by lower domestic opportunity cost.

## Terms of Trade Rule

Mutually beneficial exchange requires the Terms of Trade ($TOT$) to settle strictly between both domestic opportunity cost ratios:

$$\left|\frac{\Delta Y}{\Delta X}\right|_{\text{Region A}} < TOT < \left|\frac{\Delta Y}{\Delta X}\right|_{\text{Region B}} \tag{2}$$

Specialization and trade shift the Consumption Possibilities Frontier ($CPF$) strictly outside the domestic autarky PPF.